

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated **multi-cloud GPU clusters** (AWS, GCP, CoreWeave) with **NVIDIA/CUDA** hardware and **InfiniBand/RoCE** networking for long-running training and **high-availability inference**.
- Wrote systems-level code in **Go, Rust**, and **C++** for **scheduling, resource allocation**, and **fault-tolerant placement** under GPU-scarcity constraints.
- Built observability and **autoscaling** for **vLLM** serving via **Prometheus/Grafana**; ran distributed training via **Slurm** HPC scheduling, keeping utilization up through node loss.
- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Optimized production LLM inference using **TensorRT-LLM** alongside **vLLM**, tuning serving configuration for **low-latency, high-throughput** generation.
- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** migrating synchronous serving to async **vLLM** on **GCP Cloud Run** with autoscaling.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and SQL-backed **distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized failure-handling contracts across **25+ financial services**, reducing production escalations **60%** in one quarter via consistent **retry, timeout, and propagation** semantics.
- Standardized API contracts and failure handling across **25+ services**, reducing escalations **60%**; drove **root cause analysis** and **post-incident reviews** to improve long-term reliability.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP-compatible** server.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Go, Rust, C++, SQL, HTML/CSS

AI Infra: Model Serving, vLLM, TensorRT-LLM, Triton Inference Server, GPU Profiling, Quantization

Infra: Kubernetes, Docker, AWS, GCP, NVIDIA/CUDA, Prometheus/Grafana

Practices: Autoscaling, SLOs, Distributed Systems, Performance Optimization, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer